

Dynamic and detail-oriented mechanical engineer with over 7 years of hands-on experience in the Aerospace and Defense industries, specializing in airborne avionics, defense electronics, marine systems, spacecraft, and ground support systems. Expertise in mechanical design, part modeling, sheet metal design, and plastic molding, with a proven ability to create precise part and assembly drawings. Experience in New Product design & development, CAD/CAE, Structural and thermal analysis, CFD analysis, Tolerance stack-up, DFM, Material Selection, product benchmarking. Core competency: Thermal analysis, CFD, thermodynamics, Strength of material, Structural Analysis & TOM.

 kannan6645@gmail.com

 91-8870977752

 Bengaluru, India

 <https://www.linkedin.com/in/kannan-t-a77792121/>

WORK EXPERIENCE

Mechanical systems design & analysis engineer

Boeing - Bangalore

09/2023 - Present,

Achievements/Tasks

- Design, Testing, Development, and Qualification of Galley Equipment for Aircraft Avionics & Galley Systems.
- CFD and Thermal Analysis of Systems to optimize heat transfer Path using Flotherm and FloEFD. (steady state & transient)
- Performed modal, sinusoidal, random and shock analysis to identify high deflection areas and determine stiffening requirements using HyperMesh, Nastran, and Patran.
- Heater coil design and characterization for transient air recirculating & reheating system.
- Executed comprehensive requirements drafting with **DOORS**, ensuring 100% technical accuracy and alignment with project milestones.
- Perform hand calculation of welded and bolted joints. Electronics board design & cable routing, Hydraulic design & routing, HMI & Keypad interface design and Sensor selection.
- Optimized project transparency and stakeholder reporting by integrating **MPP** and **Smartsheet** for collaborative dashboards, ensuring 100% visibility into milestone completion.

Mechanical Design engineer

Data Patterns IND Ltd. - Chennai

02/2019 - 09/2023,

Achievements/Tasks

- Proficient in PCB mechanical design, including defining mounting constraints for both rigid and flexible PCBs through DXF and EMN (IDF) exchanges.
- Collaborates extensively with PCB layout and hardware engineers on component placement, considering thermal and structural factors.
- Extensive experience in designing LEO satellites, including spacecraft and ground station antennas, as well as airborne and defense products such as military and weather radar systems, while adhering to MIL-STD standards.
- Skilled in designing LRU chassis according to ARINC standards (both air- and conduction-cooled), IP-rated enclosures, RF and digital modules (COTS), sheet metal & plastic enclosures, ground support systems, test jigs, and standard 19" racks.
- Component in torque calculations for servo selection, thermal and structural hand calculations, stack-up analysis, hydraulic pressure drop calculations, and selecting heat sinks and cooling fans.
- Experienced in selecting shock mounts, thermal pads, and Wedge-Lok components, and knowledgeable about various instruments such as chillers, flow equipment and sensors..
- Knowledgeable in material selection, finishing processes, and manufacturing techniques.
- Extensive experience in thermal analysis of products using FloTHERM and FLOEFD software.
- Performed FEA simulation of structural analysis using Hyper Mesh, NASTRAN, and PATRAN software.
- Design simulation for structural and thermal analysis using Ansys. Perform Computational flow dynamics (CFD) analysis to optimize heat accumulation inside product.
- Experienced in overall PLM of defense avionics systems, including technical discussions, vendor communication, and documentation for preliminary and critical design reviews.
- Competency: Tolerance stack-up, GD&T, CAD, CAE, Moldflow analysis, DFMA, DFMEA, Supplier co-ordination, RCA.

EDUCATION

Bachelor of technology, Mechanical Engineering

SASTRA University, Trichy

2015 - 2019,

86%

INTERESTS

- Trekking
- Adventure
- Travelling

LANGUAGES

- Tamil
- English
- Hindi

ACHIEVEMENTS

SASTRA Racing Team:

Lead designer of Motor and power transmission systems for Electric Car Vehicle Championship (ESVC) conducted by ISIE. Recorded 4th place for the best Electric car design in overall Asia.

CERTIFICATIONS

- *Kinematics: Describing the Motions of Spacecraft – Coursera certificate*
- *Rocket propulsion – NPTEL certificate*

SKILLS

- CFD Analysis
- CATIA V5
- Thermal analysis
- Flotherm
- FloEFD
- PTC Creo
- Altair Hypermesh
- SolidWorks 2021
- Analytical skill
- Structural analysis
- Patran 2022
- Nastran 2022
- Moldex 3D
- Tolerance stack-up (WC, RSS) GD &T
- 3D Modelling
- AutoCAD

PROJECTS

Design and Development of Micro Satellite

- Lead Mechanical Design Engineer of a Micro Satellite focused on agricultural and remote sensing applications.
- **Learning:** Space technologies such as sun sensors, gyroscopic sensors, magnet torquers, reaction wheels, propulsion thrusters, control moment gyros, and multi-layer insulation (MLI) sheets.
- Executed dynamic load computations to ensure effective attitude control and stability, structural rigidity, and stable thermal integrity of the satellite.

VHF Radar Design for Aircraft Monitoring:

- Developed a 15m x 7m Yagi antenna radar system on a military truck for detecting stealth aircraft and drones, with 10 RPM azimuth rotation.
- Motor Torque calculations, gearbox selection, and antenna deployment (2.5m to 15m).
- Integrated electro-mechanical ball screw actuators for vertical elevation (0° to 90°).

Design & Development of Electronic Warfare Systems for Fighter Aircraft

- Radar Warning Receiver (**RWR**) and Airborne Early Warning and Control (**AEW&C**) system, designed for various fighter platforms including **LCA Tejas** and **Sukhoi-30** as per Arinc ATR standard.
- **Challenges:** fan selection and placement for effective cooling. Utilized forced convection with two 120 CFM airborne fans and ECS chilled air (10°C at 18 CFM) for managing 600W power dissipation. Brazed fins were added over card guides for improved thermal management. Also focused on shock mount selection for resilience in airborne conditions.

Design and Development of 3U,6U VPX/VME Modules, Customized RF Modules

- Designed many RF, power and Digital modules for 3U and 6U VPX form factor based on VITA 48.2 standard.
- **Learning:** Mechanical PCB design considerations (Components placement, Mask opening, heat tower), Proper shielding with Nolato gasket for EMI/EMC compatibility.

Design of Maritime Patrol Radar (MPAR)

- MPAR is a versatile AESA based maritime surveillance airborne radar system for search and rescue missions for Dornier Aircraft platform.
- **Learning:** The overall electronics dissipates around 1.8kW at heat sink. Heat sink is analyzed for various fin configurations (Straight, Staggered and Elliptical fins) and Fan positions to achieve efficient heat transfer at an ambient of 55°C.
- Selection of Servo motor along with gearbox by considering overall torque requirement, Pointing accuracy and operation speed (60 RPM).

Design & Structural Analysis of Aircraft Cabin Interior Systems

- Led the design, testing, development, and qualification of galley equipment's for aircraft.
- FEA analysis for ensuring structural rigidity.
- Thermal analysis for cooling of electronic control systems.
- Selection of appropriate actuators and sensors for system functionality.
- Hydraulic design and routing optimization to minimize pressure loss.
- Electrical cable routing to facilitate easy assembly/disassembly and reduce EMI interference.
- HMI and keypad design focused on aesthetics and usability.
- CFD analysis for optimizing heat and airflow distribution.
- Conducted **modal/ random / shock analysis** to determine the modes and max. stresses area using **Hypermesh, Nastran, and Patran** as per the input from **DO-160**.
- Completed Thermal testing of system at ambient of -55°C to +85°C ambient to verify the storage and functional requirements as per **D6-36075**.
- Performed reliability / cyclic testing for COTS and complete system based on complete mission profiles of 777, 737 and 787 Aircrafts.